A CASE-STUDY BASED REPORT

ON

Forest Fire Risk Prediction Using Random Forest and Gradient-Boosted Trees

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INDEX

Section	Page
Abstract	3
Introduction	4
Background	4
Motivation for the Project	4
Overview of Work Carried Out	4
Scope and Responsible-Use Statement	4
Problem Statement	5
Case-Study Scenario	5
Central Case-Study Question	5
Formal Problem Definition	5
Objectives that Guide the Study	5
Assumptions and Limitations	5
Dataset and Exploratory Data Analysis	6
About the Dataset	6
Data Preparation and Cleaning	6
EDA Exploration - Implementation	7
Summary of EDA Findings	10
Algorithm Selection and Implementation	11
Random Forest and Gradient Boosting	11
Evaluation and Cross-Validation	12
Model Interpretation	13
Final Prediction System and Results	14
Conclusion	15

Abstract

Forest fires threaten ecosystems, public safety, property and critical infrastructure, making early risk assessment important for proactive disaster management. This case study develops a binary machine-learning classifier that labels observed conditions as Fire or Not Fire using meteorological variables and Fire Weather Index (FWI) components. The supplied Algerian Forest Fires dataset was cleaned to 244 usable observations. The final modelling table used 12 predictors after excluding the constant Year field and encoding the target class. Random Forest and Gradient Boosting were trained on an 80/20 stratified split and evaluated with Accuracy, Precision, Recall, F1-Score and ROC-AUC. Random Forest classified all 49 observations in the selected test split correctly and achieved 97.94% mean accuracy under 5-fold stratified cross-validation. Gradient Boosting achieved 95.92% test accuracy and 97.14% mean cross-validation accuracy. Feature analysis showed that Initial Spread Index (ISI), Fine Fuel Moisture Code (FFMC) and Fire Weather Index (FWI) were among the strongest signals. The final Random Forest model was saved and connected to a simple prediction interface that accepts environmental and FWI readings and returns a Fire/Not Fire classification with a model-estimated probability.

Keywords: forest fire risk, Random Forest, Gradient Boosting, Fire Weather Index, binary classification, ensemble learning, early warning.

Introduction

Background

Forest-fire risk depends on interacting weather and fuel conditions rather than on a single measurement. Temperature, relative humidity, rainfall and wind influence fuel dryness and fire spread, while FWI components such as FFMC, DMC, DC, ISI, BUI and FWI summarise different aspects of moisture, drought, buildup and spread potential. Because these relationships are non-linear, ensemble decision-tree models are suitable for learning the pattern that separates fire from not-fire conditions.

Motivation for the Project

A forest-management team may receive many weather and fire-index readings every day. Manually combining all of them can be slow and inconsistent. The project therefore asks whether a machine-learning model can use the recorded conditions as a decision-support tool to flag risky situations early, helping teams prioritise monitoring and preparedness.

Overview of the Work Carried Out

The dataset was structured and cleaned, invalid/header rows were removed, one malformed record was corrected, numeric fields were validated, and potential outliers were reviewed rather than automatically discarded. EDA was then used to study class balance, distributions, class-wise differences and correlations. Random Forest and Gradient Boosting classifiers were trained, compared on a held-out test set, checked with 5-fold stratified cross-validation, interpreted with feature importance, and assessed with ROC curves. The better-performing Random Forest model was saved for a prediction prototype.

Scope and Responsible-Use Statement

This is a small academic case study based on one regional dataset. The output is a model estimate, not an emergency warning or a guarantee that a real fire will or will not occur. Several predictors are themselves FWI-derived fire-weather indicators, so high predictive performance should not be interpreted as proof that raw weather alone can forecast ignition. In operational use, predictions should supplement—not replace—expert judgement, local observations and official fire-management procedures.

Problem Statement

Case-Study Scenario

Illustrative operational case built around the supplied dataset. A forest division monitors weather during the dry season and records temperature, humidity, wind, rainfall and FWI components. The control room wants a fast screening tool that can combine these readings and indicate whether the current pattern resembles historical Fire or Not Fire conditions. The goal is not to replace field crews, but to prioritise attention when conditions resemble the fire class.

Central Case-Study Question

"Can the forest division use weather and Fire Weather Index measurements to identify high-risk fire conditions early and consistently?"

Formal Problem Definition

Given 12 input features describing calendar, weather and FWI conditions, predict a binary target $y \in \{0,1\}$, where 0 = Not Fire and 1 = Fire. The study compares Random Forest and Gradient Boosting classifiers under the same train/test split and evaluation metrics.

Objectives that Guide the Study

- Prepare a valid modelling dataset and verify missing values, duplicates, invalid ranges and suspicious outliers.
- Use EDA to understand how weather/FWI features differ between Fire and Not Fire observations.
- Train Random Forest and Gradient Boosting models using the same 80/20 stratified split.
- Compare Accuracy, Precision, Recall, F1-Score, ROC-AUC and 5-fold cross-validation accuracy.
- Interpret the selected model using feature importance and create a user-input prediction prototype.

Assumptions and Limitations

- The supplied dataset is treated as representative of the case-study environment, although it contains only 244 observations.
- All records belong to 2012; Year is therefore constant and is excluded from modelling.
- Outliers are retained when values are physically plausible because extreme weather is relevant to fire risk.
- FWI-derived features are strongly related to the target and may increase measured classification performance.
- Test-set results depend on the selected split; cross-validation is therefore reported as the stronger generalisation check.

Dataset and Exploratory Data Analysis

About the Dataset

The project uses the supplied Algerian Forest Fires dataset. After preparation, 244 observations remain with 14 original columns. The target column Classes contains two labels: Fire and Not Fire. For modelling, the constant Year column and the text target are excluded from X, while Classes is encoded as Fire_Label (Fire = 1, Not Fire = 0). This leaves 12 predictive features.

Group	Columns	Meaning
Calendar	Day, Month, Year	Observation date; Year is constant (2012).
Weather	Temperature, Relative Humidity, Wind Speed, Rain	Direct meteorological conditions.
FWI system	FFMC, DMC, DC, ISI, BUI, FWI	Fuel moisture, drought, buildup and spread/intensity indicators.
Target	Classes	Binary label: fire / not fire.

Data Preparation and Cleaning

The raw file contained formatting/header rows and inconsistent whitespace. Numerical fields were converted to numeric values, target labels were normalised, invalid/header rows were removed, and one malformed observation was corrected so that all 244 valid records were retained. The final dataset contains no missing values and no duplicate rows. Range checks found no impossible day, month, humidity, rainfall, wind-speed or negative FWI-system values.

Cleaning / validation step	Final status
Formatting and repeated-header rows	Removed
Whitespace / class-label consistency	Normalised
Malformed record	Corrected
Missing values	0 remaining
Duplicate rows	0
Invalid numerical ranges	None detected
Final records	244

Class balance: 138 Fire observations (56.6%) and 106 Not Fire observations (43.4%). The distribution is reasonably balanced, so no synthetic oversampling was applied.

Key preparation code

```
df["Fire_Label"] = df["Classes"].map({"not fire": 0, "fire": 1})
X = df.drop(columns=["Classes", "Fire_Label", "Year"])
y = df["Fire_Label"]
```

EDA Exploration - Implementation

EDA was performed with Pandas, Matplotlib and Seaborn to understand class balance, feature distributions, class-wise separation and relationships among numerical variables. The first set of plots shows the overall distribution of the target, temperature, relative humidity and Fire Weather Index.

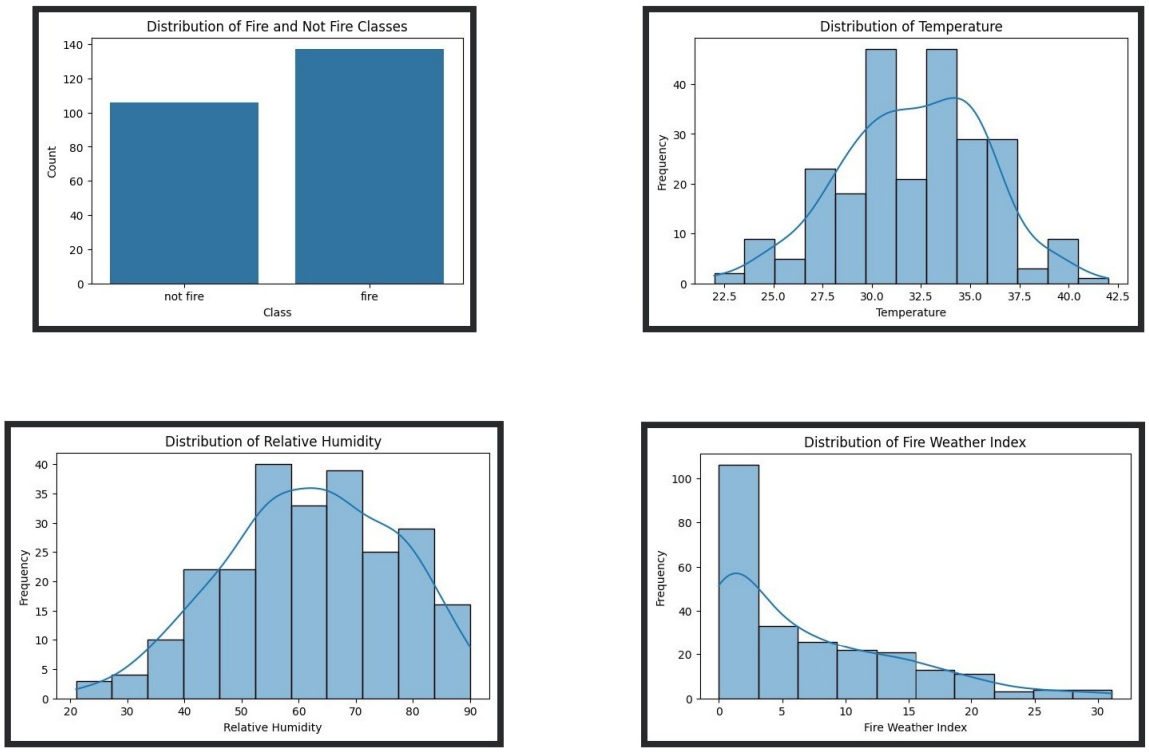


Figure 1: Class distribution and distributions of temperature, relative humidity and Fire Weather Index

- The target is reasonably balanced, with slightly more Fire cases than Not Fire cases.
- Temperature is concentrated mainly around the low-to-mid 30°C range.
- Relative humidity is widely distributed, with many observations around 50-70%.
- FWI is strongly right-skewed: many observations have low values and fewer observations have very high values.

Class-wise EDA and Correlation

The boxplots compare four important variables across the target classes. Fire observations tend to be hotter, drier and much higher in FWI, while Not Fire observations contain more rainfall and higher relative humidity.

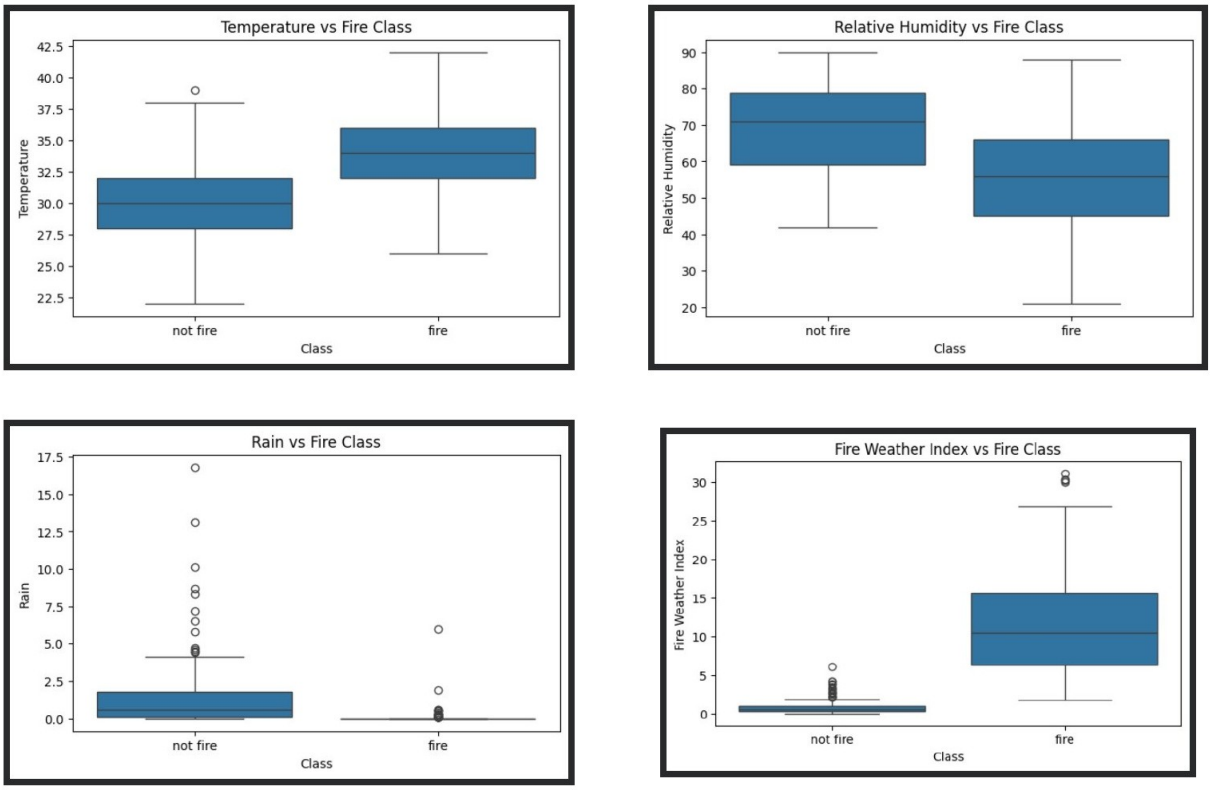


Figure 2: Temperature, relative humidity, rainfall and FWI by target class

Feature	Fire mean	Not Fire mean	Main pattern
Temperature	33.82	30.03	Higher in Fire
Relative Humidity	56.28	69.31	Lower in Fire
Rain	0.10	1.62	Much lower in Fire
ISI	7.43	1.28	Much higher in Fire
FWI	11.72	0.96	Much higher in Fire

Correlation Analysis and EDA Findings

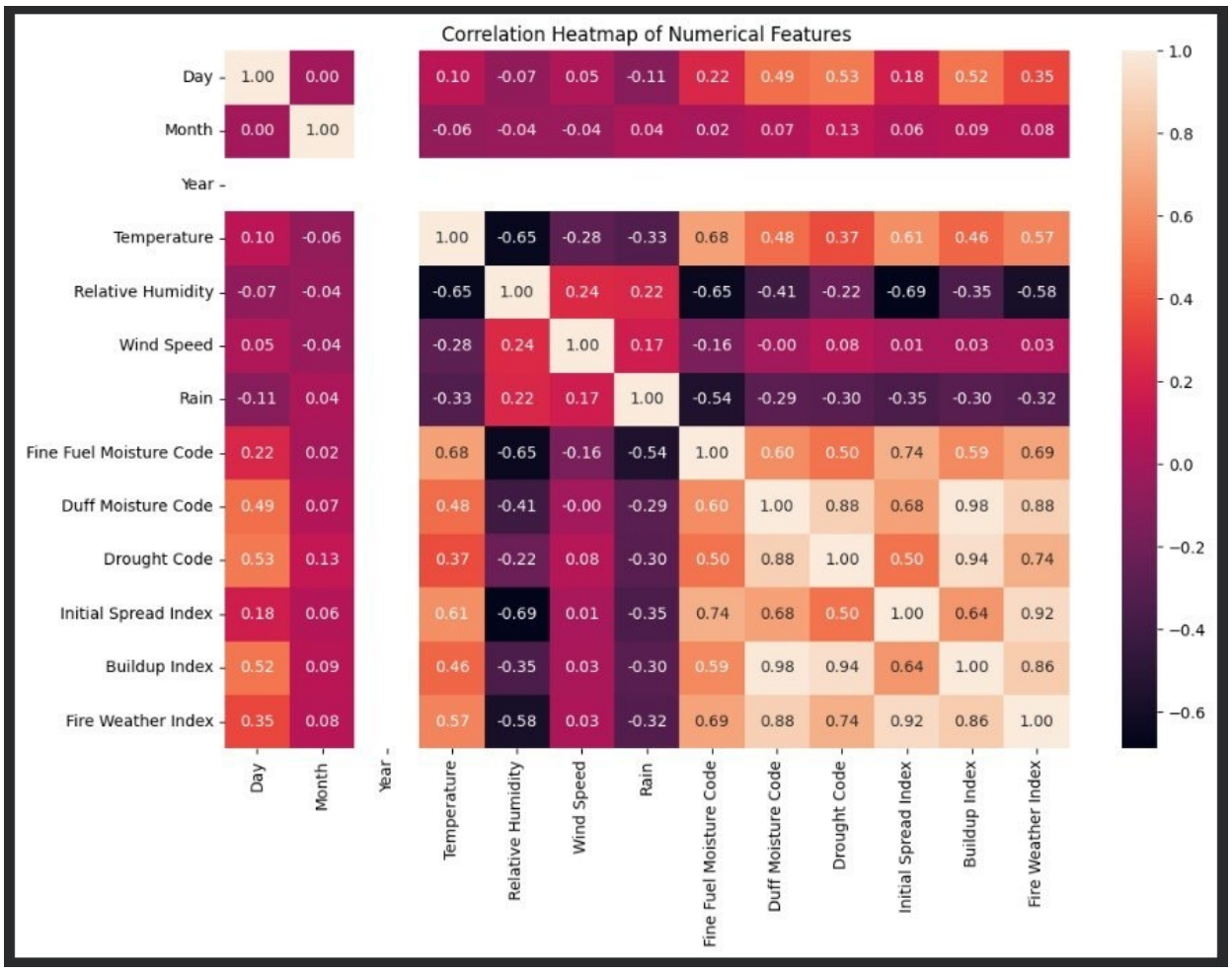


Figure 3: Correlation heatmap of numerical features

The FWI components are strongly related to one another. Notable correlations include DMC-BUI ≈ 0.98, DC-BUI ≈ 0.94, ISI-FWI ≈ 0.92, DMC-FWI ≈ 0.88 and BUI-FWI ≈ 0.86. Temperature is positively related to FWI (≈ 0.57), while relative humidity (≈ -0.58) and rain (≈ -0.32) are negatively related to FWI.

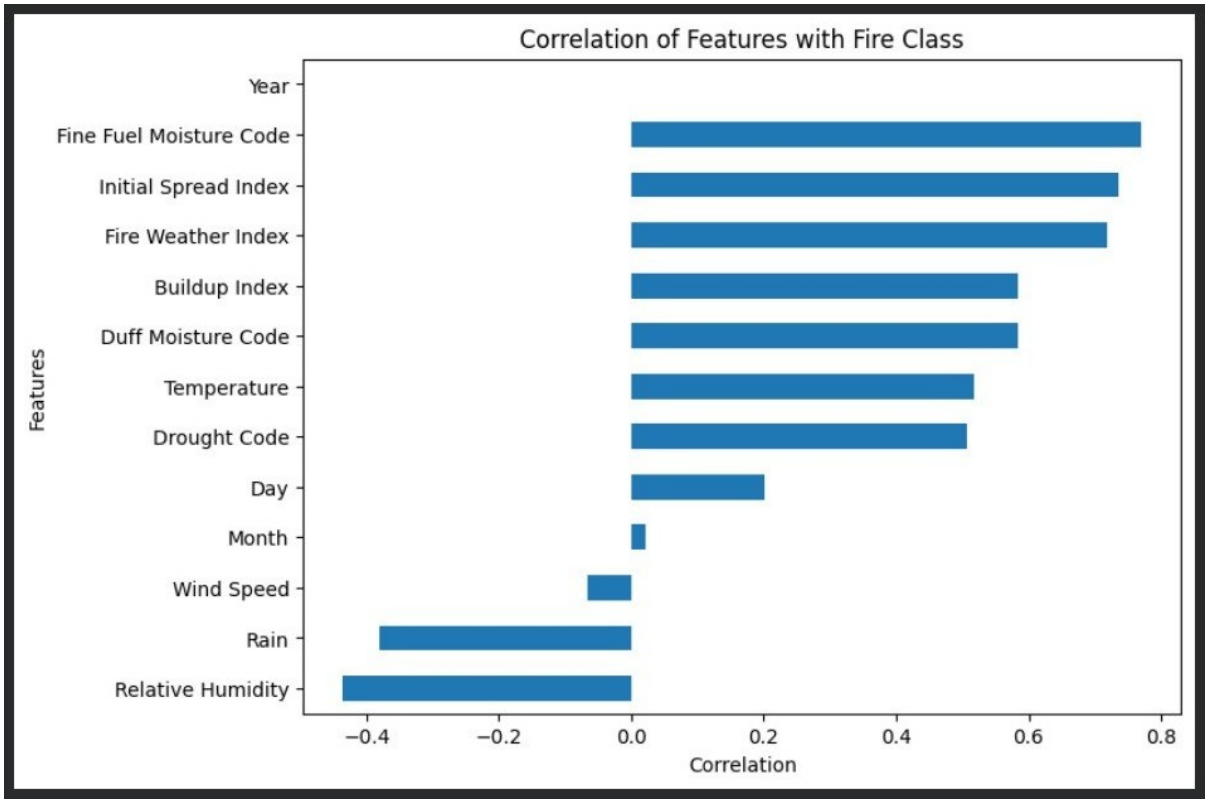


Figure 4: Correlation of numerical features with the Fire class

EDA conclusion. Fire observations are characterised by higher temperature and fire-weather indices and by lower humidity and rainfall. FFMC, ISI and FWI show strong positive association with the Fire label, while relative humidity and rain show negative association. Because multiple features overlap and interact, tree-based ensemble models are appropriate for classification.

Algorithm Selection and Implementation

Algorithm Selection

The project compares two ensemble tree classifiers. Random Forest uses bagging: many decision trees are trained on bootstrap samples and combined to reduce variance. Gradient Boosting builds trees sequentially, with each new tree focusing on errors made by the current ensemble. Both algorithms can model non-linear thresholds and interactions without requiring feature scaling.

Model	Main settings	Reason for use
Random Forest	n_estimators=100, random_state=42	Robust bagged baseline; handles nonlinear feature interactions.
Gradient Boosting	n_estimators=100, learning_rate=0.1, max_depth=3, random_state=42	Sequentially refines errors and provides a strong ensemble comparison.

Train/Test Split

A stratified 80/20 split was used with random_state=42. This produced 195 training observations and 49 test observations. The training set contained 110 Fire and 85 Not Fire cases; the test set contained 28 Fire and 21 Not Fire cases. Stratification preserved the target ratio in both subsets.

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)

rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
gb_model = GradientBoostingClassifier(
    n_estimators=100, learning_rate=0.1, max_depth=3, random_state=42
)
rf_model.fit(X_train, y_train)
gb_model.fit(X_train, y_train)
```

Held-Out Test Results

Metric	Random Forest	Gradient Boosting
Accuracy	1.0000	0.9592
Precision	1.0000	0.9333
Recall	1.0000	1.0000
F1-Score	1.0000	0.9655
ROC-AUC	1.0000	0.9991

Evaluation, Cross-Validation and Model Interpretation

On the selected test split, Random Forest classified all 49 observations correctly. Gradient Boosting classified 47 of 49 correctly: both errors were false positives (Not Fire predicted as Fire), while all 28 Fire observations were detected. Because a single split can be optimistic on a small dataset, 5-fold stratified cross-validation was also used.

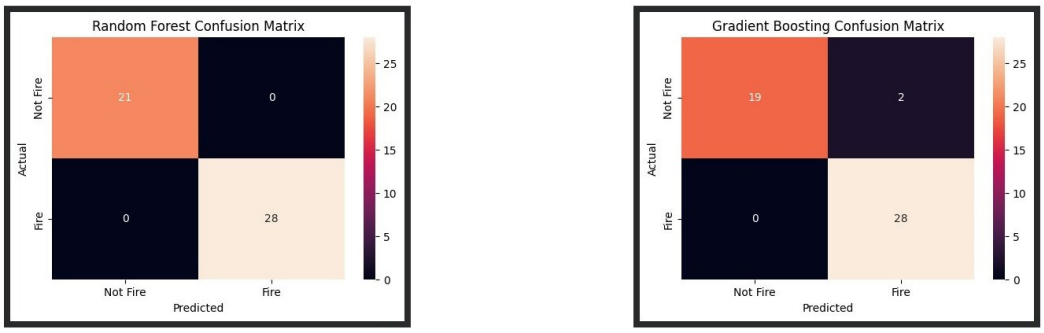


Figure 5: Confusion matrices for Random Forest (left) and Gradient Boosting (right)

5-fold cross-validation results

Model	Fold accuracies	Mean	Std. dev.
Random Forest	0.9592, 1.0000, 0.9796, 1.0000, 0.9583	0.9794	0.0184
Gradient Boosting	0.9388, 0.9796, 0.9796, 0.9592, 1.0000	0.9714	0.0208

Random Forest therefore has the slightly higher mean cross-validation accuracy (97.94% vs 97.14%) and slightly lower variation across folds. This supports selecting Random Forest as the final model while still recognising that both methods perform strongly on this dataset.

Feature Importance and Model Comparison

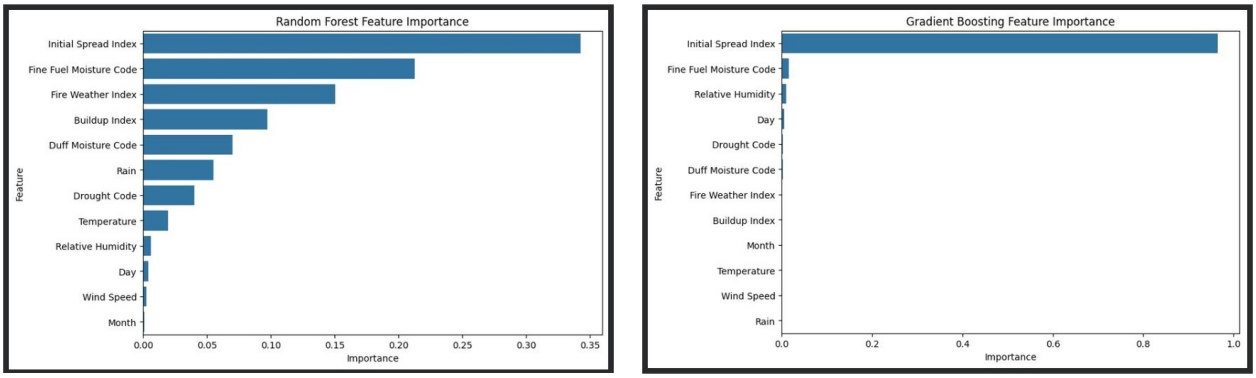


Figure 6: Feature importance from Random Forest (left) and Gradient Boosting (right)

Random Forest distributes importance across several fire-weather variables: ISI ≈ 34.2%, FFMC ≈ 21.3%, FWI ≈ 15.0%, BUI ≈ 9.7% and DMC ≈ 7.0%. Gradient Boosting concentrates most impurity-based importance on ISI (≈ 96.4%). This does not mean the remaining variables are irrelevant; strong correlation among FWI-system variables allows a boosted tree to rely heavily on one of several overlapping signals.

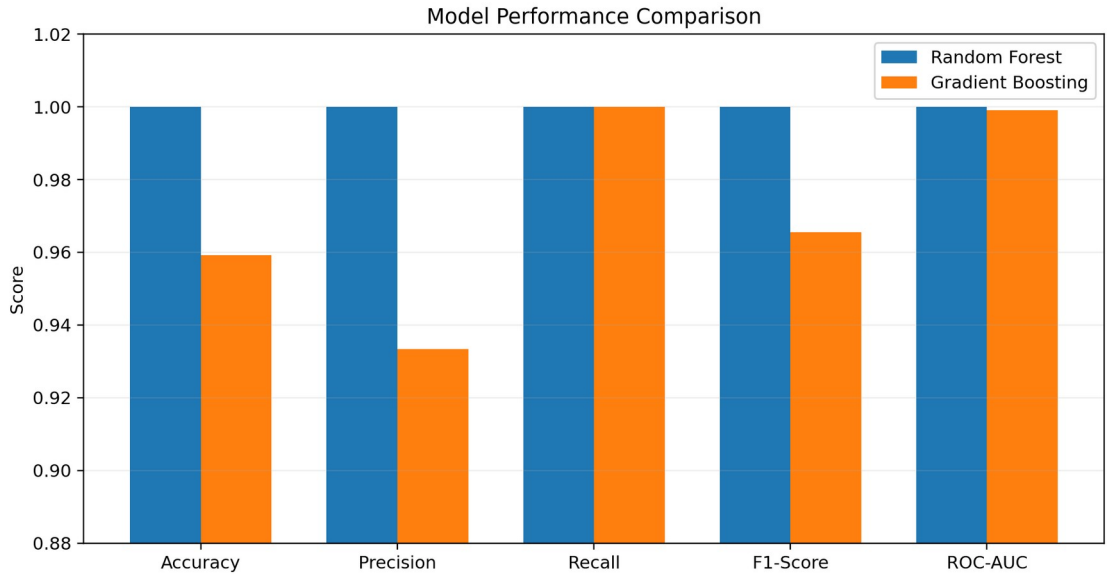


Figure 7: Held-out model performance comparison

Final Prediction System and Results

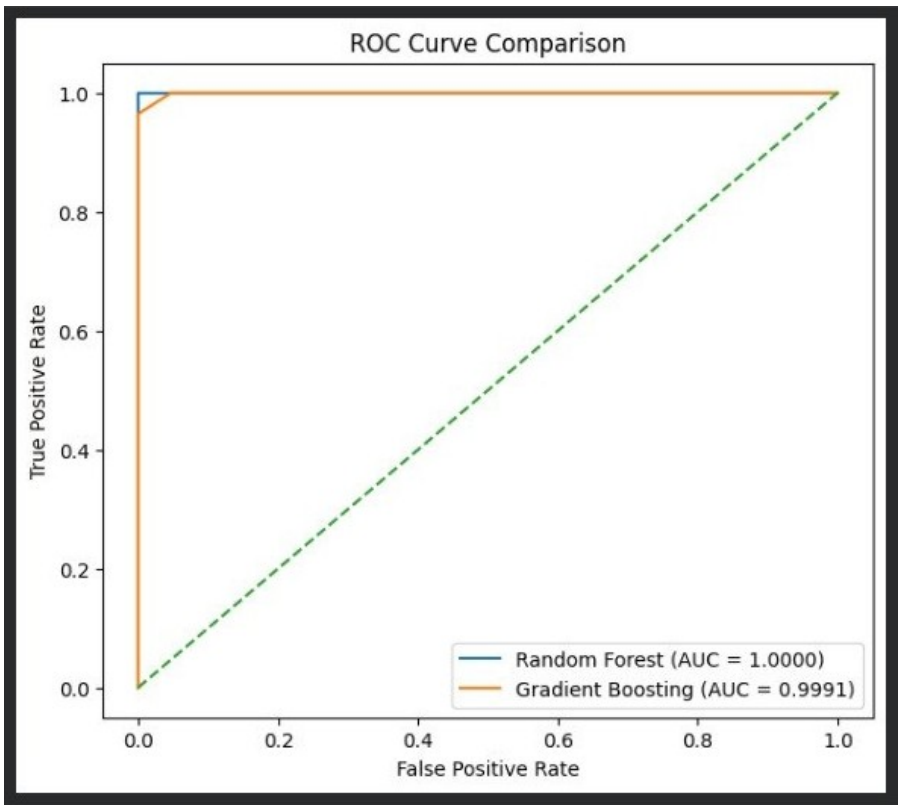


Figure 8: ROC curve comparison on the held-out test set

Both ROC curves lie very close to the top-left corner. Random Forest obtained AUC = 1.0000 and Gradient Boosting AUC = 0.9991 on the selected test set. Combined with cross-validation, Random Forest was retained as the final model. The selected model was then retrained on all 244 observations and saved with Joblib for use in a prediction prototype.

Prototype Workflow

1. User enters Day, Month, weather variables and six FWI-system values.
2. The frontend sends the values to a Python prediction service.
3. The saved Random Forest model returns class 0/1 and predict_proba().
4. The interface displays NOT FIRE / FIRE and the model-estimated fire probability.

Demonstration case	Representative inputs	Prediction	Model output
Low-risk example	Temp 27, RH 80, Rain 2.5, FFMC 55, ISI 0.5, FWI 0.4	NOT FIRE	0% in the demonstrated run
High-risk example	Temp 35, RH 45, Rain 0, FFMC 90, ISI 10, FWI 15	FIRE	100% in the demonstrated run

Important interpretation: predict_proba() is a model-estimated probability based on tree votes, not a guaranteed real-world probability that a fire will occur. The final system should be presented as an academic decision-support prototype.

Conclusion

This case study built a complete forest-fire risk classification workflow from data preparation through deployment. The supplied Algerian Forest Fires dataset was cleaned to 244 valid observations, explored with EDA, encoded for binary classification, and modelled using Random Forest and Gradient Boosting. Fire cases were associated with higher temperature and fire-weather indices and with lower relative humidity and rainfall.

Random Forest achieved 100% accuracy on the selected 49-observation test split and 97.94% mean accuracy in 5-fold stratified cross-validation. Gradient Boosting achieved 95.92% test accuracy and 97.14% mean cross-validation accuracy. Both models achieved 100% recall for the Fire class on the selected test split, but Random Forest produced fewer errors overall and was selected as the final model.

Feature importance and EDA consistently highlighted ISI, FFMC, FWI and related fuel/drought indicators as strong signals. The saved Random Forest model was connected to a user-input prototype so that environmental readings can be converted into a Fire/Not Fire prediction with a model-estimated probability.

Limitations and Future Work

- The dataset is small and covers a limited geographic/time setting, so broader external validation is needed.
- FWI-derived predictors are closely related to the target and may make the classification task easier than raw-weather-only forecasting.
- Future work can test the model on additional regions/seasons, calibrate predicted probabilities, tune hyperparameters, and compare with XGBoost/LightGBM.
- For operational use, the system should integrate current sensor/weather feeds and official fire-management thresholds, with human review.

Signature of the Team Members

Signature of the Guide